Follow the Workers, Not the Bids: Cartel Detection through Employer–Employee Networks*

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Abstract

Brazil's antitrust authority convicted 35 firms across seven procurement cartels in São Paulo between 2009 and 2019. We link these firms to 2.6 million competitive auctions on the state's electronic platform and to the universe of matched employer–employee records, assembling what is, to our knowledge, the first detection benchmark to combine auction behavior, labor microdata, and formal antitrust rulings.

The benchmark exposes a sharp divide. On roughly 2,300 close-margin auctions involving at least one convicted firm, canonical variance screens behave as bid-rotation theory predicts—dispersion and bidder counts are compressed—yet the areas under the ROC curve (0.52–0.56) are too low to be operationally useful. The close-margin RD of Kawai et al. (2023) likewise yields a null. Worker-flow screens tell a different story: counting the employees who have moved between the winning and runner-up firms produces AUCs of 0.75 and 0.74, outperforming every classical screen by roughly 20 percentage points. A co-bidding density measure reaches 0.70. Crucially, the worker-flow advantage is not uniform: in thin labor markets the gap reaches 33 percentage points; in thick markets it narrows to 10. A specialization placebo shows that about half the signal reflects the natural personnel circulation of thin markets, while the other half is a cartel-specific residual.

The pattern points to an institutional logic. Where specialized-labor pools are shallow, the cheapest way to coordinate a cartel runs through people, not through prices—and the traces show up in employment records that canonical bid screens never see.

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For competition authorities with access to employer–employee microdata, worker-flow screens offer a practical, low-cost complement to the toolkit they already have.

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1. Introduction

Do the canonical tools for detecting procurement cartels work when they leave the thick-market, high-enforcement environments in which they were built? We put this question to a new test. Linking seven cartels convicted by Brazil's antitrust authority to 2.6 million competitive auctions and the full universe of formal employment relationships in São Paulo, we assemble the first detection benchmark that combines auction data, employer–employee records, and criminal ground truth. Classical variance screens and a close-margin regression discontinuity yield AUCs of 0.52–0.56—directionally right but operationally inadequate. Screens built from inter-firm worker flows reach 0.75, and the gap widens sharply in thin labor markets where collusion coordinates through people rather than prices. This paper documents these facts, traces them to an institutional mechanism, and draws out the consequences for enforcement design.

A growing literature uses statistical regularities in bidding behavior to infer collusion that is never documented directly. Classical screens exploit the compression that bid-rotation agreements impose on the bid distribution: prearranged prices cluster around the designated winner, yielding low dispersion, few active bidders, and predictable rotation (Porter and Zona, 1993; Pesendorfer, 2000; Bajari and Ye, 2003; Conley and Decarolis, 2016; Imhof et al., 2018). The complementary approach of Kawai et al. (2023) focuses on the winning–losing threshold: under genuine competition, firms that barely win and barely lose should look alike, and any sharp discontinuity at the cutoff identifies a non-competitive wedge. Both families of tests have been validated primarily against cartels in high-income economies with deep bidder pools and strong institutional capacity for enforcement; their track record in thinner settings rests on a small and growing empirical base.

Network-based methods have pushed the frontier in two directions. Within the procurement data itself, co-bidding networks flag candidate cartels through group-level cohesion and exclusivity (Wachs and Kertész, 2019; Signor et al., 2020; Huber and Imhof, 2019; Clark et al., 2018). Outside the bid record, a recent wave exploits external network layers: Asai and Charoenwong (2026) show that common-ownership links predict bid rigging in Singapore, and Villamil et al. (2024) build multiplex ownership–co-bidding networks for Swedish procurement. What remains missing is a network layer that captures *operational* coordination—the personnel flows, knowledge transfers, and shared specialized labor through which cartels in thin markets actually run—and a validation of such a

layer against criminal ground truth in a middle-income setting. The São Paulo–CADE–RAIS linkage we construct offers precisely that test, and is the first cartel-detection benchmark to integrate auction-level data, matched employer–employee records, and formal antitrust convictions. The policy stakes are immediate: middle-income competition authorities increasingly rely on screens developed and validated abroad to triage investigations (Organisation for Economic Co-operation and Development, 2015; Bosio et al., 2022), and if those screens systematically miss cartels under local conditions, enforcement pipelines may be misfiring where deterrence is needed most.

Our primary evaluation sample is a *cartel-active* benchmark of roughly 2,300 close-margin pregão auctions in which at least one convicted firm bid during the active period of its cartel (details of sample construction in Section 5).

On that benchmark, canonical screens disappoint. Bid dispersion and the coefficient of variation are weakly compressed at cartel-tainted auctions (Cohen's d between -0.07 and -0.24), and Welch t-statistics are significant thanks to the enormous control sample—but the AUCs top out at 0.56, far short of what a triage tool would require. The close-margin RD produces a null on the cartel subsample. In short, the screens pick up a whisper of signal, but not enough to act on.

Worker-flow screens recover what bid screens cannot. Counting the employees who have worked at both the winning and the runner-up firm over the 2009–2017 panel yields an AUC of 0.75; the Jaccard similarity of the two firms' workforces reaches 0.74; a co-bidding density measure adds 0.70. A specialization placebo decomposes the worker-flow signal into a same-market component—reflecting the natural labor circulation of thin product markets—and a cartel-specific residual. Roughly half the signal comes from each channel, consistent with cartels both inheriting and reinforcing the personnel flows already present in the markets where they operate.

We read these findings through the lens of *institutional thinness*. When the universe of firms capable of bidding on a given product is small and the formal-sector labor pool that serves those firms is shallow, coordination through personnel becomes the cheapest technology available to a cartel. The bid distribution, in such markets, carries little trace of the collusion—but the employment records do. A simple model in Section 4.6 formalizes this logic and delivers three testable predictions: classical screens should be weakest in thin markets; worker-flow screens should be strongest there; and a same-market placebo should absorb part, but not all, of the worker-flow signal. All three predictions are borne out. The sharpest confirmation comes from splitting the bench-

mark by market thickness: the gap between the best classical screen and the worker-flow screen jumps from 10 percentage points in thick markets to 33 percentage points in thin ones.

The paper makes three contributions. The first is conceptual. Collusion leaves *institution-specific* signatures: the screen that detects a cartel depends on which coordination technology is cheapest in the relevant environment. In thick labor markets, collusion channels through bids and classical screens fire; in thin ones, it channels through people and worker-flow screens fire. This framing organizes—and partially reconciles—the contradictory findings that have accumulated as detection methods developed in one setting are deployed in another (Conley and Decarolis, 2016; Imhof et al., 2018; Kawai et al., 2023). The second contribution is substantive. We build conviction-anchored benchmarks by linking CADE rulings, BEC bid-level records, and RAIS employer–employee data, and show that worker-flow and co-bidding screens constructed from these sources substantially outperform classical screens. Because matched employer–employee registries are increasingly available to national competition authorities in middle-income economies, the screens we propose are directly deployable at negligible marginal cost. The third contribution is methodological: we deliver the first stress test of canonical detection frameworks against criminal ground truth outside a high-income setting, documenting that classical screens and the Kawai RD yield AUCs in the low 0.50s while network screens reach 0.75.

The remainder of the paper proceeds as follows. Section 3 describes public procurement in São Paulo and the CADE cartel cases that anchor the ground truth. Section 2 situates our work in the detection, network, and worker-mobility literatures. Section 5 describes the data sources and sample construction. Section 4 presents the empirical strategy and a simple model of institution-dependent coordination. Section 6 reports the results. Section 7 discusses policy implications and concludes.

2. Related Literature

Our paper sits at the intersection of three literatures: the design and validation of cartel-detection screens, the use of network structure to identify collusion, and the economics of inter-firm worker mobility.

2.1. Detection through bid-pattern screens

The idea that bid-rotation agreements leave statistical footprints is now well established. Porter and Zona (1993) showed that collusive phases of the 1880s railroad freight cartel exhibit systematically lower bid dispersion than competitive phases, exploiting the cartel's own internal records as an exogenous regime switch. Pesendorfer (2000) documented bid rotation in Florida school-milk procurement; Bajari and Ye (2003) proposed exchangeability tests for bidder costs in Midwestern highway paving; Conley and Decarolis (2016) extended pair-level correlation tests to Italian procurement. Imhof et al. (2018) demonstrated that combining multiple screens in a portfolio outperforms any single indicator in Swiss data, and Abrantes-Metz et al. (2006) offered practical guidance on implementing variance screens while cautioning against single-indicator reliance. Harrington (2008) synthesized the design principles, emphasizing a point our findings will make concrete: the informativeness of any screen depends on the competitive counterfactual *and* the coordination technology the cartel actually uses.

A complementary strand focuses on the bid margin itself. Kawai et al. (2023) proposed a regression discontinuity centered on the winning–losing threshold in Japanese construction procurement: under genuine competition, narrowly winning and narrowly losing firms should look alike, and any sharp discontinuity identifies a non-competitive wedge. Chassang and Ortner (2019) developed a theoretical framework in which bid constraints endogenously shape the collusive technology—predicting, among other things, that format-specific screens may outperform generic variance tests, a prediction consistent with the format heterogeneity we document below.

A common thread across these contributions is that nearly all validation exercises rely on ground truth from high-income economies with deep bidder pools and well-resourced enforcement agencies. Levenstein and Suslow (2006) documented across 81 international cartels that market structure, transparency, and enforcement capacity jointly determine cartel success—suggesting that screens validated in one institutional regime need not travel well to another. Direct validation against criminal ground truth in a middle-income setting with structurally different labor markets remains scarce, and that is the gap our conviction-anchored benchmark is designed to fill.

2.2. Network-based and machine-learning approaches

A parallel literature has moved beyond univariate bid statistics to exploit the relational structure of procurement markets. Wachs and Kertész (2019) showed that co-

bidding networks in Georgian procurement flag candidate cartels through group-level cohesion and exclusivity, even without criminal ground truth. Signor et al. (2020) built a probabilistic detector from the data released by Brazil's *Operação Lava Jato*, reporting 81–96 percent accuracy on the Petrobras construction cartel. Huber and Imhof (2019) combined classical screens with supervised machine-learning classifiers in Swiss data, correctly predicting 84 percent of collusive tenders out of sample. Clark et al. (2018) documented how collusive networks in Québec construction co-evolve with political connections, illustrating the interplay between bid rigging and corruption.

A more recent wave extends the network logic to data sources *outside* the bid record. Asai and Charoenwong (2026) link shareholder registries to procurement auctions in Singapore and show that common-ownership links predict identical bids; removing the ownership-network effect would cut government spending by 4–6 percent. Villamil et al. (2024) build temporal multiplex networks that combine ownership ties and co-bidding patterns in Swedish procurement, finding that denser ownership connections predict higher single-bidding and more missing bidders. Together, these papers establish that external network layers—corporate ownership, in their case—carry detection signal that co-bidding statistics alone cannot reach.

We add a third layer: labor-market connectivity, constructed from matched employer–employee records. Worker-flow networks differ from ownership networks in two ways that matter for detection. They capture *operational* coordination—personnel rotation, knowledge transfer, shared technical staff—rather than financial ties, and thus speak to a distinct coordination channel. And they are available wherever governments maintain compulsory employer–employee registries (RAIS in Brazil, DADS in France, IAB in Germany), making the approach portable without new data collection. Because worker-flow data are informationally independent of both the bid record and the corporate registry, the risk of circularity between the screen and the behavior it targets is lower than for co-bidding or ownership-based approaches.

2.3. Worker mobility as an inter-firm channel

Labor economists have long recognized that workers carry knowledge between firms. Stoyanov and Zubanov (2012) showed that productivity gains spill across Danish firms through the mobility of their employees, with larger spillovers when the productivity gap between sender and receiver is wide. Serafinelli (2019) documented a similar pattern in Italy's Veneto region: workers arriving from high-productivity firms raise wages and

output at their new employers. Jarosch et al. (2021) developed an equilibrium model in which team composition and worker reallocation jointly shape firm-level human capital. The upshot is that worker flows are not just a labor-market phenomenon—they are an active conduit for inter-firm information transmission.

Three recent papers connect this conduit to collusion. Lamichhane and McGee (2024) provide experimental evidence that, in Bertrand markets, worker mobility between firms facilitates collusive pricing: workers carry pricing-strategy information, and firms that exchange employees coordinate on higher prices and more market sharing, especially in concentrated markets. Herrera-Caicedo et al. (2025) study product-market collusion facilitated by shared directors and executives. They find that common leadership raises the probability of collusion by 11 percentage points—and, strikingly, that inter-firm worker flows *fall* by roughly 20% once a board interlock is established. The substitution is telling: when collusion runs through leadership ties, personnel-flow coordination becomes redundant. Our institutional-thinness model predicts the mirror image: when leadership ties are absent and coordination relies on operational personnel exchange, worker-flow screens should carry the signal that ownership and board-interlock screens miss.

On the data-infrastructure side, two papers have already linked RAIS to Brazilian procurement—though not for detection. Barbosa and Straub (2020) track worker movements between public agencies and private suppliers, showing that supplier-to-administration revolving doors are detrimental to procurement efficiency. Szerman (2023) exploits a debarment-policy change to estimate the worker-level costs of anti-corruption enforcement, finding that blacklisted firms lose nearly half their employees. Both papers prove the RAIS–procurement linkage is feasible and productive; neither uses it as a cartel screen.

We bring these threads together. Lamichhane and McGee (2024) establish the mechanism—worker mobility as a collusion channel. Asai and Charoenwong (2026) and Villamil et al. (2024) show that external network layers carry detection signal. Barbosa and Straub (2020) and Szerman (2023) demonstrate the data infrastructure. What has been missing is the step from mechanism to screen: constructing a labor-network detection tool, validating it against formal convictions, and embedding it in a framework that predicts *when* worker-flow screens should outperform the alternatives. That is what this paper provides.

3. Institutional Background

3.1. Public procurement in São Paulo: the BEC platform

São Paulo's state government purchases common goods and services through a decentralized network of roughly 1,300 public buyer units (PBUs), all of which have been required since 2007 to transact on the state's electronic platform, the *Bolsa Eletrônica de Compras* (BEC). The scale is substantial: in 2018 alone, BEC handled approximately US$1.7 billion in procurement, and over the study period the platform processed more than R$7.4 billion for the State Health Secretariat and other agencies.

The BEC catalog organizes items by group and class—from pharmaceuticals to office supplies—totaling about 6,350 distinct product codes. Each tender, called a purchase offer (PO), may contain multiple items; each item is auctioned separately, so a unique procurement event is defined by a (PO $\times$ item code) pair, which we label POI. The platform records everything a researcher needs: the reference price set by the buyer, every bid submitted, the identities of bidders and the winning firm, and the auction format.

Two formats predominate. In the *convite* (sealed-bid tender), firms submit confidential bids by a deadline; the lowest qualified bid wins. This format is typically used for purchases below R$176,000 (about US$35,000). In the *pregão eletrônico* (descending auction), firms first submit sealed proposals, after which an open phase begins: bidders observe the current lowest price in real time and can underbid for an initial 20 minutes, with automatic 4-minute extensions whenever a late bid arrives. The auction closes after 4 consecutive minutes of silence. The pregão carries no value ceiling and accounts for the majority of BEC spending.

The distinction matters for collusion. The pregão's iterative, anonymous bidding gives cartel members the ability to monitor compliance in real time but also exposes deviations; the convite's single-shot format makes monitoring harder but bid coordination simpler. Our primary analysis focuses on pregão auctions, for which the close-margin RD framework is most naturally motivated; convite results are reported as a complementary check.

3.2. Antitrust enforcement: CADE and the ground truth

Brazil's federal competition authority, the *Conselho Administrativo de Defesa Econômica* (CADE), is an autonomous agency within the Ministry of Justice. Its Tribunal renders

administrative decisions—binding fines and behavioral remedies, not criminal sentences in the judicial sense. Cases typically begin with a complaint or a leniency application (*Termo de Compromisso de Cessação*, TCC), proceed through a formal investigation by the *Superintendência-Geral*, and conclude with a Tribunal vote.

Our ground truth draws on CADE proceedings in which the Tribunal convicted firms of cartel conduct in São Paulo procurement markets. We searched CADE's public repository for proceedings with final decisions that (i) name specific firms with identifiable CNPJs, (ii) specify the product market, and (iii) delineate a conduct period. Seven cartels meet these criteria, spanning sectors as varied as pharmaceuticals, school meals, trash bags, solar water heaters, and metropolitan rail, and involving 35 firms whose CNPJ roots we can match to BEC records.

What makes these case files particularly useful is that they specify not just *who* colluded but *when* and *in which market*. We exploit this granularity to label individual auctions as cartel-tainted—matching each convicted firm's CNPJ root, the CADE-specified conduct window, and the relevant product code to BEC tenders. The result is a benchmark that captures both the identity and timing dimensions of collusion, rather than treating every auction involving a flagged firm as equally suspect.

4. Empirical Strategy

We bring four detection approaches to the conviction-anchored benchmark: a close-margin regression discontinuity (Section 4.1), classical bid-distribution screens (Section 4.2), network screens built from worker-flow and co-bidding data (Section 4.3), and a specialization placebo that separates market-level labor circulation from cartel-specific coordination (Section 4.5). Every screen is evaluated by its area under the ROC curve (Section 4.4). A simple model in Section 4.6 ties the pieces together.

4.1. Close-margin regression discontinuity

Following Kawai et al. (2023), we focus on auctions decided by a hair. Define the margin of victory in auction a as

$$MV_a = \frac{b_a^{\text{runner-up}} - b_a^{\text{win}}}{b_a^{\text{win}}}.$$

When MV_a is close to zero, the winner barely underbid the nearest rival. Under genuine competition, narrowly winning and narrowly losing firms should look alike on predetermined characteristics; a sharp discontinuity at the cutoff signals something non-competitive.

We estimate the jump using local-polynomial regression with a data-driven bandwidth. Two outcomes carry interpretive weight. *Incumbency*—whether the firm won the previous auction in the same market—tests for bid rotation: a jump would indicate that designated winners are systematically past winners. *Last bidder*—whether the firm submitted the final bid—tests for strategic timing. A density test on the running variable checks whether the margin itself is being manipulated.

4.2. Classical bid-distribution screens

We compute four screens on firm-best bids (one bid per firm per auction, keeping the lowest when multiple rounds exist):

1. **Coefficient of variation** (CV_a): standard deviation divided by mean. Bid rotation predicts lower CV because cover bids cluster near the designated winner (Porter and Zona, 1993; Abrantes-Metz et al., 2006).
2. **Bid spread** (S_a): $(\max - \min)/\text{mean}$. Like CV, compressed under cover bidding.
3. **Margin of victory** (MV_a): the designated winner should undercut cover bids by just enough to win, yielding a small margin.
4. **Number of bidders** (N_a): entry deterrence or market division should reduce the count (Marshall and Marx, 2012).

For each, we compare cartel-tainted auctions to clean controls on the close-margin benchmark ($|MV| \leq 0.10$), reporting the Welch t-statistic, Cohen's d, the Wilcoxon rank-sum p-value, and the AUC.

4.3. Network-based screens

We step outside the bid data entirely and build three screens from administrative sources that CADE analysts do not routinely examine.

Worker-flow screens. For each (winner, runner-up) pair we measure labor connectivity from the full 2009–2017 RAIS panel:

- **Shared workers** (W_{ij}): the count of distinct PIS numbers appearing at both firms during the nine-year window.

- **Jaccard similarity** ($J_{ij} = W_{ij}/(N_i + N_j - W_{ij})$): normalizes the count by firm size, so that large employers are not mechanically flagged.

The logic is straightforward. If cartel coordination in thin labor markets runs through personnel—managers carrying pricing information, technicians rotating between member firms—then the employment records should show it (Stoyanov and Zubanov, 2012; Serafinelli, 2019).

Co-bidding screen. For each pair we count the auctions in which both firms submitted bids (C_{ij}). This provides a network channel built entirely within the procurement data, serving as a benchmark against which the value-added of the external worker-flow layer can be measured.

4.4. Detection evaluation: AUC

We use the area under the ROC curve to compare screens. The AUC is simply the probability that a randomly chosen cartel auction scores more "cartel-like" than a randomly chosen clean auction. For screens where lower is more suspicious (CV, S, MV, N), we negate scores so that AUC > 0.5 always means detection in the predicted direction. We compute the AUC via the Mann–Whitney U statistic with average-rank tie-handling:

$$\text{AUC} = \frac{U_1}{n_1 \times n_0},$$

where n_1 and n_0 are the cartel and clean sample sizes. All screens are evaluated on the same unified close-margin benchmark, so comparisons are apples-to-apples. Appendix A.1 shows that results are stable across bandwidths from 0.02 to 0.50.

4.5. Specialization placebo

Elevated worker flows between a cartel firm and its rival could mean two very different things: that the firms collude through personnel, or that they simply draw from the same thin labor pool. We need to tell them apart.

We build a three-way decomposition. The **cartel** sample consists of auction-level (winner, runner-up) pairs in which at least one firm is cartel-active ($N \approx 2{,}300$ for pregão). The **placebo** consists of non-cartel pairs bidding on the same BEC item code in the same year—firms facing the same product-market labor pool but not colluding. The **clean** baseline is every other non-cartel pair.

If the entire worker-flow signal were market specialization, the placebo should look like the cartel sample. If there is a cartel-specific coordination channel on top, cartel pairs should show still more worker overlap than same-market non-cartel pairs. We measure the split by comparing AUCs: the fraction of the total gain (cartel vs. clean, minus 0.5) absorbed by the placebo is the specialization share; the remainder is the cartel residual.

4.6. A simple model of institution-dependent coordination

Why should screen effectiveness depend on the market? Consider $n \geq 2$ firms that can form a cartel. They need to coordinate on a winning bid for each auction. Two technologies are available:

- **Bid-pattern coordination** (cost c_b): arrange cover bids around the designated winner's price. Leaves traces in CV, spread, and bidder counts—exactly what classical screens are designed to catch.
- **Personnel-flow coordination** (cost $c_w(L)$, decreasing in the thickness L of the specialized-labor pool): coordinate through transfers, rotations, or shared employment of key staff. Leaves traces in worker-flow statistics but not necessarily in the bid distribution.

Rational cartel members pick the cheaper channel. In thick markets (large L), $c_w > c_b$: collusion runs through bids, and classical screens fire. In thin markets (small L), $c_w < c_b$: collusion runs through people, and the bid distribution looks clean.

Three testable predictions follow:

1. **Thickness gradient.** In thin markets, classical AUCs should be near chance while worker-flow AUCs should be above chance. In thick markets, the gap should narrow as classical screens recover.
2. **Placebo decomposition.** Thin labor pools produce spontaneous inter-firm worker circulation even without collusion. A same-market placebo should show elevated worker overlap relative to random pairs (specialization), and cartel pairs should show more still (coordination). Both components should be present.
3. **Symmetry.** Because personnel-flow coordination operates through the shared labor pool of the market—not through the outcome of any particular auction—the worker-flow signal should not depend on whether the cartel firm wins or loses.

The model is deliberately spare. It abstracts from repeated-game dynamics, leniency, the possibility that firms use both channels simultaneously, and heterogeneity in c_b across auction formats. We offer it not as a structure to be estimated but as a lens for reading the results and for specifying when the classical-vs.-network comparison is most informative.

5. Data

The analysis rests on three administrative data sources—procurement bids, employment records, and antitrust rulings—which we link at the firm level to construct the conviction-anchored benchmark.

5.1. BEC: bid-level procurement data

We draw on the complete bid-level records from São Paulo's *Bolsa Eletrônica de Compras* (BEC) for January 2009 through December 2019. After restricting to pregão and convite procedures with at least two participating firms, the data contain 2,613,010 competitive auctions, 13,957,419 firm-level bid participations, 40,273 distinct firms, and 153,048 unique item codes. Each record identifies the purchase offer, item code, participating firm (by CNPJ), bid amount, outcome, auction format, and purchasing agency.

For each auction we compute the screens described in Section 4.2—coefficient of variation, bid spread, margin of victory, and bidder count—on firm-best bids (keeping each firm's lowest bid when multiple rounds exist). Markets are defined transitively following Kawai et al. (2023): two firms share a market if they ever bid on the same item code in the same auction. This yields 7,581 distinct markets.

5.2. RAIS: employer–employee matched panel

The *Relação Anual de Informações Sociais* (RAIS) is Brazil's compulsory annual census of formal employment. Every establishment with a CNPJ must report each employment bond active at any point during the year, identified by the worker's PIS/PASEP number. We use the 2009–2017 vintages, restricting to establishments whose 8-digit CNPJ root appears in the BEC firm universe—roughly 19,000 roots. The result is a nine-year panel of every formal-sector job at every BEC-active firm.

From this panel we build the worker-flow edge list. For each pair of BEC firms (i, j), we count distinct PIS numbers that appear at both firms at any point during the nine

years (the shared-worker count W_{ij}) and compute the Jaccard similarity J_{ij}. Pairs with no shared workers are zero-imputed when merged to the auction benchmark.

5.3. CADE: cartel ground truth

Our ground truth comes from CADE administrative proceedings with final Tribunal decisions convicting named firms of cartel conduct in São Paulo procurement markets (see Section 3). Table 1 lists the seven cartels we identified—spanning pharmaceuticals, school meals, trash bags, solar water heaters, metropolitan rail, airport cafeterias, and school transport—together with their conduct periods and the 35 firms whose CNPJ roots match BEC records.

The key feature for detection is that each case file specifies not just the colluding firms but the product market and time window. We exploit this to label individual auctions as cartel-tainted: an auction earns the flag only if a convicted firm bid during the CADE-specified conduct period for the relevant product and year. This firm × period × product matching avoids the coarser alternative of treating every auction involving a flagged firm as suspicious.

Table 1: Convicted Cartels in São Paulo Procurement Markets

Sector	CADE process	Period	Firms	In BEC	Conduct type
Trens e metrôs	08700.004617	1999–2013	13	10	Bid rigging
Merenda escolar	08012.010022	2006–2013	7	6	Market division
Medicamentos	08012.002222	2007–2011	8	8	Hub-and-spoke
Sacos de lixo	08700.005789	2008–2014	3	3	Bid rigging / price fixing
Aquecedores solares	08012.001273	2009–2010	5	4	Bid rigging
Cafeterias Infraero	08700.007278	2014	6	3	Bid rigging
Transporte escolar	08700.005876	2019	2	2	Bid rigging
Total (unique CNPJ roots)			35	36	

Notes: "Firms" and "In BEC" count unique 8-digit CNPJ roots per cartel; row totals are after cross-cartel deduplication. The In BEC total exceeds the Firms total because one firm participates in two cartels. Conduct periods hand-curated from CADE Tribunal decisions and press releases (see Appendix). Source: script 12.

5.4. Sample construction

All detection tests use the close-margin envelope $|MV| \leq 0.10$. Within that envelope we define three samples:

1. **Cartel-active benchmark** ($N \approx$ 2,300 pregão; $N = 263$ convite). At least one convicted firm bid during the active period of its cartel. This is the primary evaluation sample.
2. **Strict pair-matched sample** ($N = 10$ pregão). Both the winner and the runner-up belong to the same convicted cartel during the same conduct period. The gold standard for head-to-head coordination, but power-limited.
3. **Anytime-listed sensitivity** ($N \approx 80$ pregão). Both firms appear in a CADE conviction for the same cartel, but the auction year may fall outside the conduct window. Of the 107 pre-filtering auctions, 93 are out-of-window and 100 come from a single case (the medicamentos hub-and-spoke, CADE 08012.002222/2011-09). Reported as a sensitivity only.

The *clean control* is every close-margin auction with no cartel-listed firm ($N =$ 527,259 pregão; 519,557 convite). All screens are evaluated against this common baseline.

5.5. Variable definitions

Table 2 collects the key variables.

Table 2: Variable Definitions

Variable	Definition
MV_a	Margin of victory: $(b_a^{\text{runner-up}} - b_a^{\text{win}})/b_a^{\text{win}}$
CV_a	Coefficient of variation of firm-best bids
S_a	Bid spread: $(\max - \min)/\text{mean}$ of firm-best bids
N_a	Number of distinct firms submitting bids
Incumbent_a	$= 1$ if the winning firm won the preceding auction in the same market
Last bidder_a	$= 1$ if the winning firm submitted the final bid
W_{ij}	Shared workers: distinct PIS appearing at both firms i and j
J_{ij}	Jaccard similarity: $W_{ij}/(N_i + N_j - W_{ij})$
C_{ij}	Co-bidding count: auctions where both i and j bid

6. Results

We present the empirical results in three parts. Section 6.1 reports the performance of classical bid-distribution screens and the close-margin RD on the unified benchmark.

Section 6.2 reports the performance of worker-flow and co-bidding network screens. Section 6.3 reports the specialization-placebo decomposition.

6.1. Classical screens and the close-margin RD

Table 6 presents the unified benchmark comparison of all screens on the same close-margin pregão sample ($|MV| \leq 0.10$). Panel A reports the classical bid-distribution screens on the cartel-active benchmark (N = 2,319 cartel-tainted auctions vs. N = 527,259 clean controls).

All four classical screens fire in the direction predicted by the bid-rotation model: the coefficient of variation ($\Delta = -0.035$, $d = -0.07$), bid spread ($\Delta = -0.173$, $d = -0.10$), and number of active bidders ($\Delta = -0.87$, $d = -0.24$) are compressed at cartel-tainted auctions relative to clean controls. Welch t-statistics are large and statistically significant, owing to the massive control sample. Yet the effect sizes are small: Cohen's d ranges from -0.07 to -0.24, indicating near-total distributional overlap between cartel and clean auctions. The AUCs reflect this overlap, ranging from 0.52 (coefficient of variation) to 0.56 (number of bidders)—above chance but far too low for investigative triage. A Wilcoxon rank-sum test confirms the weak rank-order separation ($p < 10^{-4}$ for each screen, consistent with the AUC direction).

The margin of victory exhibits an anomalous pattern: $\Delta MV = +0.004$ ($d = +0.17$), with an AUC of 0.46—slightly *below* chance and in the *opposite* direction to the bid-rotation prediction. Cartel-active auctions have marginally *larger* winning margins, not smaller. We interpret this as consistent with a partial phantom-bidding mechanism in which cover bids are inflated relative to the designated winner's price, widening rather than compressing the margin. This finding warrants further investigation but does not affect the headline conclusion: classical screens as a portfolio do not achieve operational discriminative power on this benchmark.

A direct replication of the Kawai et al. (2023) close-margin RD detects a baseline incumbency advantage on the full sample ($\hat{\beta} = 0.018$, SE $= 0.005$, $p < 0.001$), reflecting a competitive pattern in which recent winners are slightly more likely to win again. On the cartel-active subsample, however, no *additional* discontinuity is detected ($p > 0.15$): cartel exposure does not amplify the baseline incumbency pattern, nor does it produce a detectable jump in last-bid behavior. A density test in the spirit of Imhof et al. (2018) detects no manipulation of the running variable on the cartel-active benchmark

($T_{\text{CJM}} = -0.02$, $p = 0.98$). Neither the RD design nor the density test flags the cartel-active auctions as anomalous relative to competitive bidding.

6.2. Network-based screens

If classical screens cannot separate cartel from clean on this benchmark, what can? Panel B of Table 6 turns to the network screens, evaluated on the same $N = 2{,}319$ cartel-active auctions.

The answer is worker flows. The shared-worker count yields an AUC of 0.750 ($\Delta = +3.70$, $t = +14.1$), and the Jaccard similarity yields an AUC of 0.739 ($\Delta = +0.0014$, $d = +0.19$, $t = +20.7$). Both outperform the best classical screen (number of bidders, AUC = 0.56) by roughly 20 percentage points. A third screen constructed from co-bidding density yields an AUC of 0.695, providing independent corroboration through a distinct information channel.

Critically, the worker-flow signal is not an artifact of sectoral composition. Restricting attention to firm pairs that share the same 2-digit CNAE sectoral classification reduces the Jaccard AUC only marginally, from 0.739 to approximately 0.730. The residual of worker-flow after partialling out the log co-bidding density of each pair achieves an AUC of 0.749—essentially unchanged from the raw baseline, indicating that worker mobility carries detection information above and beyond what co-bidding patterns alone can reveal.

The pattern holds across auction formats. On the convite subsample ($N = 263$ cartel-active, $|MV| \leq 0.10$), worker-flow AUCs remain strong (0.704 and 0.701), even as classical screens lose direction entirely. That worker-flow screens are format-invariant while classical screens are format-sensitive is itself consistent with the institutional-thinness mechanism: bid distributions depend on auction rules, but the personnel connections that sustain coordination do not.

These headline numbers raise a natural question. In most cartel-active auctions, only *one* of the two firms in the (winner, runner-up) pair is a convicted cartelist—so what exactly is the AUC measuring? Of the 2,319 cartel-active auctions, 996 have the *winner* as the cartel-active firm and the runner-up as a non-cartel firm; 1,313 have the *runner-up* as the cartel-active firm; and only 10 have both firms cartel-active. Decomposing the AUC by cartel role clarifies the interpretation:

Subgroup	N	AUC (shared workers)	AUC (Jaccard)
Winner cartel, runner-up clean	996	0.741	0.730
Runner-up cartel, winner clean	1,313	0.759	0.747
Both cartel-active	10	0.558	0.550

Three observations emerge. *First*, the AUC is remarkably stable across the two one-sided subgroups (0.74–0.76), indicating that the detection signal is symmetric: it does not matter whether the cartel firm wins or loses the auction. This rules out the concern that the screen merely captures a "winner effect" driven by a cartel firm's incumbency or market dominance. *Second*, the signal is present even when the *runner-up* is the cartel firm (AUC = 0.759), confirming that elevated worker-flow connectivity is a property of cartel-firm pairs in thin labor markets, not an artifact of winning per se. *Third*, intra-cartel pairs (both firms cartel-active, $N = 10$) exhibit the highest mean shared-worker count ($\bar{W} = 28.4$, compared with 6.5 for winner-only and 2.1 for clean controls) but an AUC near chance (0.56), reflecting severe power constraints at small N rather than an absence of signal.

The decomposition clarifies the mechanism: the worker-flow screen detects that cartel firms are embedded in thin specialized-labor markets where personnel flows with *all* rivals—not just co-conspirators—are dense. The specialization placebo (Section 6.3) further separates this market-embeddedness component from the cartel-specific coordination residual. The headline AUC of 0.75 thus captures both channels; the cartel-specific residual, net of market specialization, is approximately 0.64 (Section 6.3).

6.3. Specialization-placebo decomposition

The worker-flow signal could reflect two quite different forces: the natural personnel circulation that thin product markets generate among *all* firms, or the incremental coordination that cartels layer on top. To separate them, we build a three-way comparison (Table 7).

The raw match rates are revealing. Cartel pairs share at least one worker in 57.9% of cases; same-market non-cartel pairs share one in 31.1%; pairs from other markets, in just 9.9%. The Jaccard AUC against the global clean baseline is 0.740 for cartel pairs and 0.605 for same-market placebo pairs. In a linear decomposition of the total AUC gain (cartel vs. clean minus 0.5):

- The **specialization component** (placebo vs. clean) absorbs approximately 0.105 AUC points, or roughly 44% of the total signal.
- The **cartel-specific residual** (cartel vs. placebo) accounts for approximately 0.138 AUC points, or roughly 56% of the total signal.

Both components are consistent with the institutional-thinness mechanism of Section 4.6: thin specialized-labor markets generate spontaneous labor circulation among same-market firms (the specialization component), and cartels additionally reinforce those flows to sustain coordination (the cartel-specific residual). The two-component structure also has a practical implication: a worker-flow screen deployed without conditioning on market identity would capture both components, potentially flagging non-colluding same-market firm pairs as suspicious. Conditioning on market or product code absorbs the specialization component, yielding a purer cartel signal at the cost of a modest reduction in overall AUC (from 0.740 to 0.638).

6.4. Heterogeneity by market thickness

The placebo tells us what part of the signal is cartel-specific. The next question is where that signal lives. If the institutional-thinness mechanism of Section 4.6 is right, worker-flow screens should outperform classical screens most sharply in thin markets—and classical screens should work better, perhaps even well, in thick ones. We test this by splitting the benchmark according to market thickness, measured as the median number of bidders per auction for each item code in the clean control sample. Markets with a control-sample median at or below the overall median of 5 firms are classified as "thin"; those above are classified as "thick." This proxy is constructed entirely from the non-cartel sample to avoid contamination by cartel-induced changes in bidder counts.

Table 3 reports the results. The pattern is striking. In thin markets ($N_{\text{cartel}} = 1{,}618$), classical screens fire in the *wrong* direction: CV AUC = 0.45, spread AUC = 0.44, and bidder-count AUC = 0.40—all below chance. Cartel-tainted auctions in thin markets exhibit *more* bid dispersion and *more* active bidders than clean controls, consistent with a phantom-bidding or cover-bid mechanism in which all cartel members participate and submit inflated bids. By contrast, in thick markets ($N_{\text{cartel}} = 416$), classical screens fire in the predicted direction: CV AUC = 0.59, spread = 0.60, bidder count = 0.60—substantively above chance and consistent with the standard bid-rotation model.

Worker-flow screens show the opposite gradient. In thin markets, shared-worker AUC reaches 0.774 and Jaccard AUC reaches 0.758; in thick markets, both decline to approximately 0.70. The gap between the best classical screen and the worker-flow screen is 33 percentage points in thin markets but only 10 percentage points in thick markets—a three-fold difference that confirms the model's core prediction.

The tercile decomposition (reported in the notes to Table 3) reveals additional structure: the worker-flow AUC peaks in mid-thickness markets (tercile 2: AUC = 0.80) rather than in the thinnest markets (tercile 1: AUC = 0.75). This non-monotonicity is consistent with the specialization-placebo finding of Section 6.3: in the very thinnest markets, spontaneous same-market labor circulation is so dense that it partially compresses the cartel-vs-clean AUC from above, even as the cartel-specific signal remains strong.

Table 3: Screen Performance by Market Thickness

	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
Thin (≤ 5 firms)	1,618	217,662	0.447	0.436	0.400	**0.774**	**0.758**
Thick (> 5 firms)	416	224,074	0.594	0.604	0.604	0.699	0.694
All (benchmark)	2,319	527,259	0.523	0.531	0.555	**0.750**	**0.739**

Notes: Market thickness is the median number of bidders per auction for each item code, computed on clean-control auctions with ≥ 5 observations. "Thin" = median ≤ 5; "Thick" = median > 5. Classical screen AUCs use "lower is cartel" convention. Tercile decomposition: T1 (median = 4): $N_t = 1{,}191$, shared-workers AUC = 0.751, CV AUC = 0.382; T2 (median = 5): $N_t = 602$, shared-workers AUC = 0.797, CV AUC = 0.533; T3 (median = 8): $N_t = 241$, shared-workers AUC = 0.644, CV AUC = 0.612.

6.5. Model predictions vs. empirical findings

Table 4 summarizes the correspondence between the three testable predictions of the institutional-thinness model (Section 4.6) and the empirical results documented in this section.

All three predictions are borne out in the data. The model's qualitative framework—that the cheapest coordination technology determines which screen family carries the detection signal—is thus not merely an organizing device but a tested empirical proposition, at least within the institutional setting of São Paulo procurement.

6.6. Selection and the interpretation of the benchmark

A natural concern with any conviction-anchored benchmark is that the cartels CADE detected and convicted are not a random sample of all cartels that existed during the

Table 4: Model Predictions and Empirical Results

Prediction	Empirical test	Result
P1. In thin markets, classical AUCs ≈ 0.5; worker-flow AUCs > 0.5.	Table 3: median split by market thickness	Classical AUC = 0.40–0.45 (inverted, exceeding prediction—suggests phantom-bidding not in model); worker-flow AUC = 0.77. **Direction confirmed.**
P2. Same-market placebo exhibits elevated worker-flow relative to random pairs; cartel pairs exhibit still more.	Table 7: three-way decomposition	Specialization component = 44%; cartel residual = 56%. Both present. **Confirmed.**
P3. Worker-flow signal is symmetric: does not depend on whether the cartel firm wins or loses.	Section 6.2: role decomposition	Winner-only AUC = 0.74; runner-only AUC = 0.76. **Confirmed.**

study period. If the detection channel itself correlates with worker-flow intensity—for instance, if CADE is more likely to investigate firms when their personnel connections are visible to investigators—then the worker-flow AUC may overstate the screen's performance on the universe of undetected cartels. We address this concern in three ways.

First, we document the evidentiary basis of each of the seven convicted cartels from the public record of CADE's administrative proceedings. Table 5 summarizes the primary detection channel and the nature of the evidence that sustained each conviction. The evidence channels are heterogeneous: one case (aquecedores solares) was convicted "based exclusively on indirect evidence"—that is, bid-pattern analysis alone; one case (transporte escolar) relied on IP address coincidence during electronic bidding; one case (sacos de lixo) originated from a state-level police operation (*Operação Colludium*); and the remaining four were built from leniency confessions, procurement-document analysis, or investigation-triggered dawn raids. Crucially, *none* of the seven cases mentions matched employer–employee data, RAIS records, or inter-firm personnel connections as part of the evidentiary basis for the investigation or conviction. CADE does not, as a matter of standard practice, use worker-flow analysis as an investigative tool—the data

linkage we construct in this paper is, to our knowledge, the first of its kind. The detection channel (CADE investigations driven by complaints, leniency, police operations, and bid-pattern screens) is therefore independent of the screening channel (worker-flow connectivity measured from RAIS).

Second, we note that the worker-flow screen is constructed from an entirely external administrative data source (RAIS, held by the Ministry of Labor) that is not part of CADE's standard enforcement toolkit. Unlike co-bidding patterns—which are constructed from the same procurement data that CADE analysts routinely examine—worker-flow connectivity requires a separate data-access agreement and a non-trivial linkage procedure. The informational independence of the two channels reduces the scope for mechanical circularity.

Third, we observe that the worker-flow AUC is relatively stable across the seven cartels despite the heterogeneity in detection channels. If selection-on-detection were driving the result, we would expect the worker-flow signal to be concentrated in cases where personnel connections played a role in the investigation. Instead, the signal is present even in the aquecedores solares case—convicted on bid-pattern evidence alone—and in the transporte escolar case—detected through IP forensics. While this cross-cartel stability is suggestive rather than definitive (small N of cartels limits formal testing), it is inconsistent with a pure selection story.

These observations do not fully eliminate the selection concern. It remains possible that firms with dense worker-flow networks are easier to detect through *any* channel—for instance, because the same organizational proximity that generates personnel flows also generates careless document trails or complaints from disgruntled employees. A definitive test would require evaluating the worker-flow screen against cartels detected through channels that are *provably* orthogonal to personnel connectivity, such as pure leniency applications from firms with no prior labor-market overlap. We leave this test to future work and flag the selection caveat as the principal interpretive limitation of our benchmark design.

7. Discussion and Conclusion

This paper assembles a conviction-anchored benchmark—seven CADE-convicted procurement cartels linked to 2.6 million competitive auctions and the universe of formal employment in São Paulo—and uses it to ask which detection tools actually work. Three

Table 5: Evidence Channels in the Seven Convicted Cartels

Cartel (sector)	CADE process	Period	Primary evidence channel
Merenda escolar	08012.010022/ 2008-16	2006–2013	Procurement documents (40,000+ docs analyzed); geographic market division
Trens e metrôs	08700.004617/ 2013-41	1999–2013	Leniency and investigation; subcontracting arrangements between winners and losers
Aquecedores solares	08012.001273/ 2010-24	2009–2010	Indirect evidence only (bid-pattern analysis); no direct evidence of agreement
Transporte escolar	08700.005876/ 2019-85	2019	IP address coincidence during electronic bidding
Cafeterias Infraero	08700.007278/ 2015-17	2014	Investigation of in-person tender procedures
Sacos de lixo	08700.005789/ 2015-02	2008–2014	Police operation (*Op. Colludium*, MP Bauru); dawn raids and document seizures
Medicamentos	08012.002222/ 2011-09	2007–2011	Hub-and-spoke investigation; leniency and administrative proceedings

Notes: Evidence channels documented from CADE public press releases and Tribunal decisions (see Appendix for full citations). None of the seven cases lists matched employer–employee data, RAIS records, or inter-firm worker flows as part of the evidentiary basis.

Table 6: Unified Benchmark: All Screens on the Same Close-Margin Sample

	Pregão ($N_t = 2{,}319$)		Convite ($N_t = 263$)	
Screen	Cohen's d	AUC	Cohen's d	AUC
Panel A: Classical bid-distribution screens				
Coefficient of variation (CV)	−0.07	0.523	−0.01	0.515
Bid spread (S)	−0.10	0.531	+0.04	0.502
Margin of victory (MV)	+0.17	0.457	+0.14	0.456
Number of bidders (N)	−0.24	0.555	+0.06	0.467
Panel B: Network-based screens				
Shared workers (W_{ij})	+0.06	**0.750**	+0.47	**0.704**
Jaccard similarity (J_{ij})	+0.19	**0.739**	+0.09	**0.701**
Co-bidding density (C_{ij})	+0.14	0.695	−0.04	0.525

Notes: All screens evaluated on the unified close-margin benchmark ($|MV| \leq 0.10$) against a common clean control ($N_0 = 527{,}259$ pregão; $N_0 = 519{,}557$ convite). Classical screen AUCs use the "lower is cartel" convention; values above 0.5 indicate detection in the predicted direction. Network screen AUCs use "higher is cartel." Bold indicates AUC ≥ 0.70. Cohen's d is the standardized mean difference (pooled SD). Source: script 23.

Table 7: Specialization-Placebo Decomposition of Worker-Flow Signal

Sample	N	% with ≥ 1 shared worker	Mean Jaccard	AUC vs. clean
Cartel-active	2,321	57.9%	0.0022	0.740
Placebo (same market)	5,313	31.1%	0.0012	0.605
Clean (global)	523,414	9.9%	0.0008	—
Signal decomposition (Jaccard AUC relative to clean baseline)				
Total signal (cartel vs. clean − 0.5)				+0.240
Specialization component (placebo vs. clean)				+0.105 (44%)
Cartel-specific residual (cartel vs. placebo)				+0.138 (56%)

Notes: Pregão, $|MV| \leq 0.10$. Placebo consists of non-cartel firm pairs bidding on the same BEC item code in the same year as each cartel-active auction. AUC computed via Mann–Whitney U (Jaccard similarity, higher is cartel). The decomposition is approximate: because AUC is a rank-order statistic, the two components do not sum exactly to the total ($0.105 + 0.138 = 0.243$ vs. total 0.240). Percentages are shares of total signal. Source: v3 specialization placebo.

findings stand out.

The first is a warning. Canonical variance screens behave as bid-rotation theory predicts—dispersion and bidder counts are compressed at cartel-tainted auctions—but the effect sizes are small and the distributional overlap with clean auctions is near-total. AUCs top out at 0.56. The close-margin RD of Kawai et al. (2023) produces a null. Neither design is internally flawed; both simply lack the statistical punch to separate cartels from noise in this institutional setting.

The second finding is constructive. Worker-flow screens built from matched employer–employee records achieve AUCs of 0.75, outperforming the best classical screen by roughly 20 percentage points. A co-bidding density measure reaches 0.70. A specialization placebo attributes about half the worker-flow signal to the natural personnel circulation of thin product markets and the other half to cartel-specific coordination. The signal is symmetric across winner and runner-up roles, survives conditioning on sector and co-bidding intensity, and is stable across bandwidths.

The third finding is the one that ties the story together. The gap between classical and worker-flow screens is not constant: it widens from 10 percentage points in thick labor markets to 33 in thin ones. In thin markets, classical screens actually fire in the wrong direction—cartel auctions show *more* dispersion, not less. This gradient confirms the institutional-thinness mechanism we develop in Section 4.6: where specialized-labor pools are shallow, the cheapest coordination technology runs through personnel rather than prices, and the detection signal migrates from bid patterns to employment records.

Policy implications. For competition authorities in middle-income economies, the message is direct. Matched employer–employee registries—RAIS in Brazil, DADS in France, IAB in Germany—are increasingly available and digitized. Building a worker-flow screen on top of existing procurement data requires a one-time linkage, not a new data-collection program. The marginal cost is negligible; the detection gain, in thin markets, is large. Authorities that rely exclusively on imported variance screens are not just under-detecting cartels—they are under-detecting them precisely in the thinnest, most vulnerable markets, where deterrence would matter most.

Limitations. The principal caveat is selection. CADE convictions are not a random sample of all cartels, and although none of the seven cases in our benchmark was built on worker-flow evidence (Section 6.6), we cannot fully rule out that firms with dense personnel networks are easier to catch through any channel. A definitive test would

require ground truth from a detection process provably orthogonal to labor-market connectivity—pure leniency applications from firms with no prior worker overlap, for instance. The strict pair-matched sample ($N = 10$) is too small to draw conclusions about head-to-head coordination; larger ground-truth sets, assembled from cross-state CADE linkages, would sharpen the test. And the simple model of Section 4.6 treats bid-pattern and worker-flow coordination as substitutes; a richer framework allowing complementarity—where cartels use both channels at once—could generate additional predictions and refine the screen-design implications.

Looking ahead. The broader lesson is that collusion is not one thing. It adapts to the institutional soil it grows in, and the tools built to detect it must adapt too. A screen calibrated for thick, well-policed markets may be exactly the wrong instrument for thin ones—not because it is poorly designed, but because it is listening for the wrong signal. Following the workers, rather than the bids, is one way to recalibrate. There are likely others.

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A. Appendix

A.1. Bandwidth sensitivity

Table A.1 reports AUCs for all screens across six bandwidth choices for the close-margin envelope. The benchmark bandwidth of $|MV| \leq 0.10$ used throughout the paper is neither the most nor the least favorable for the worker-flow screens. At narrower bandwidths (0.02, 0.05), worker-flow AUCs are marginally higher (0.76, 0.75) but sample sizes are smaller; at wider bandwidths (0.20, 0.50), AUCs decline modestly (0.74, 0.72) as the sample dilutes toward the full population. Classical screen AUCs follow the same monotonic pattern. The gap between the best classical screen (number of bidders) and the worker-flow screens remains stable at approximately 18–20 percentage points across all bandwidths.

Table A.1: AUC Sensitivity to Close-Margin Bandwidth (Pregão, Cartel-Active)

$\|MV\| <$	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
0.02	989	277,038	0.546	0.558	0.580	0.758	0.747
0.05	1,700	416,582	0.539	0.548	0.565	0.760	0.749
0.10	**2,319**	**527,259**	**0.523**	**0.531**	**0.555**	**0.750**	**0.739**
0.15	2,607	573,196	0.518	0.526	0.554	0.745	0.734
0.20	2,784	602,883	0.517	0.525	0.555	0.743	0.732
0.50	3,277	693,331	0.518	0.526	0.552	0.735	0.723

Notes: Each row reports AUCs from the unified benchmark (script 23) at the indicated bandwidth. Classical screen AUCs use the "lower is cartel" convention (negated scores). Bold row indicates the benchmark bandwidth used in the main analysis. All AUCs computed via Mann–Whitney U with average-rank tie-handling.